MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 5721

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Reflex oscillator requiring external cavity

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Ec3	Er	Ib	Ic1	Ik	T (g3 sleeve)	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	mAdc	mAdc	°C	ft
Maximum:	6.8	+25	1,250	1,250	-15	20	6.0	26	160	50,000
Minimum:	5.7	---	---	---	-800	---	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Cathode:	Coated unipotential
Mounting position:	Any
Base:	Special small 4-pin of mica filled bakelite

TEST CONDITIONS:

Parameter:	Ef	Ec1	Ec2	Ec3	Er	Ib	Ic1	Ik
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	mAdc	mAdc
Test 1:	6.3	Note 1	1,000	1,000	Adj	20	Note 1	---
Test 2:	6.3	Note 1	1,250	1,250	Adj	20	Note 1	---
Test 3:	6.3	Note 2	1,000	1,000	Adj	---	Note 2	20

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	12	---					
1301	Heater current	---	---	Ef = 6.3 V	If	520	640	mA
4229	Total reflector current	4, 5	1	Er/max Po; 7.315 < F < 7,685 MHz	Ir	---	7	μAdc
4229	Reflector-leakage current	4, 5	1	Er/max Po; 7,315 < F < 7,685 MHz	Ir	---	4	μAdc
4229	Reflector-gas current	4, 5	1	Er/max Po; 7,315 < F < 7,685 MHz	Ir	---	3	μAdc
1256	Electrode current (anode)	---	1, 2	Current to g2 and g3	Ib	---	20	mA
1261	Electrode voltage (1) (control grid)	1	---	Ec2 = Ec3 = 1,000 Vdc; Ib = 20 mA	Ec1	4	18	Vdc
1256	Electrode current (1) (control grid)	1	---	Ec2 = Ec3 = 1,000 Vdc; Ib = 20 mA	Ic1	1.0	6.0	mA
4250	Power output (1)	6, 11	1	Cavity IIB	Po	100	---	mW
4213	Reflector voltage (1)	---	---		Er	-115	-50	Vdc
4223	Mechanical tuning range (1)	---	---		F	4,190	4,390	MHz
4250	Power output (2)	6, 11	3	Cavity IIB	Po	80	---	mW
4213	Reflector voltage (2)	---	---		Er	-115	-50	Vdc
4223	Mechanical tuning range (2)	---	---		F	4,190	4,390	MHz
4250	Power output (3)	6, 11	3	Cavity IV	Po	125	---	mW
4213	Reflector voltage (3)	---	---		Er	-520	-405	Vdc
4223	Mechanical tuning range (3)	---	---		F	7,315	7,655	MHz
4250	Power output (4)	6, 11	2	Cavity VII	Po	20	---	mW
4213	Reflector voltage (4)	---	---		Er	-350	-220	Vdc
4223	Mechanical tuning range (4)	---	---		F	10,515	11,065	MHz
4214	Cathode emission	7	1	Ef = 5.8 V	ΔIb	---	15	%
4027	Temperature coefficient	8	1		Coeff	-100	+100	kHz/°C

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2							
1031	High-frequency vibration	3	---		---	---	---	---
1336	Heater-cathode leakage	---	---	Ehk = ±45 Vdc	Ihk	---	100	μAdc
1261	Electrode voltage (2) (control grid)	1	---	Ec2 = Ec3 = 1,250 Vdc; Ib = 20 mA	Ec1	2	16	Vdc
1256	Electrode current (2) (control grid)	1	---	Ec2 = Ec3 = 1,250 Vdc; Ib = 20 mA	Ic1	0.5	6.0	mA
1261	Electrode voltage (3) at Ib cutoff (control grid)	---	---	Ec2 = Ec3 = 1,000 Vdc	Ec1	-23	-8	Vdc
4250	Power output (5)	6, 11	1	Cavity I	Po	180	---	mW
4213	Reflector voltage (5)	---	---		Er	-275	-195	Vdc
4223	Mechanical tuning range (5)	---	---		F	3,555	3,715	MHz
4250	Power output (6)	6, 11	1	Cavity IIA	Po	200	---	mW
4213	Reflector voltage (6)	---	---		Er	-435	-330	Vdc
4223	Mechanical tuning range (6)	---	---		F	4,190	4,390	MHz
4250	Power output (7)	6, 11	1	Cavity III	Po	55	---	mW
4213	Reflector voltage (7)	---	---		Er	-185	-125	Vdc
4223	Mechanical tuning range (7)	---	---		F	6,700	7,040	MHz
4250	Power output (8)	6, 11	1	Cavity V	Po	100	---	mW
4213	Reflector voltage (8)	---	---		Er	-625	-490	Vdc
4223	Mechanical tuning range (8)	---	---		F	8,055	8,465	MHz
4250	Power output (9)	6, 11	2	Cavity VIB	Po	20	---	mW
4213	Reflector voltage (9)	---	---		Er	-460	-350	Vdc
4223	Mechanical tuning range (9)	---	---		F	9,650	10,150	MHz
4250	Power output (10)	6, 11	2	Cavity VIA	Po	20	---	mW
4213	Reflector voltage (10)	---	---		Er	-255	-160	Vdc
4223	Mechanical tuning range (10)	---	---		F	9,650	10,150	MHz
4280	Electronic tuning range (1)	---	1	Er for 50% max Po: 4,190 ≤ F ≤ 4,390 MHz	ΔF	12	---	MHz

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued							
4280	Electronic tuning range (2)	---	1	Er/50% max Po: 7.315 < F < 7.685 MHz	ΔF	12	---	MHz
	Quality conformance inspection, part 3							
---	Life-test provisions	9	1	Group C	t	500	---	hrs
---	Life-test end point	---	---		Po	75% of initial Po	---	---
	Periodic-check test		---					
4028	Barometric pressure, reduced	10	---		---	---	---	---

NOTES:

1. Ec1 is adjusted for a total current to the interaction gap electrode (Ib) of 20 mA with the tube in oscillation.
2. Ec1 is adjusted for a total cathode current (Ik) of 20 mA with the tube in oscillation.
3. The tube shall meet all the requirements of this tube specification sheet after vibration.
4. The tube shall be oscillating during this test in the 2-3 4 cycles reflector transit mode. Under conditions of vigorous oscillation and light loading, it is possible to draw excessive reflector current (about 1 mA) for low values of Er. No damage will be caused by this current except in cases where the reflector circuit resistance will cause Er to approach or exceed 0 volts
5. With test cavity IV as per Drawing 176-JAN. Er is adjusted in the 2-3 4 cycles reflector transit mode for maximum power output into a fixed 50-ohm load with VSWR 1.25.
6. Er is adjusted for maximum power output into a fixed 50-ohm load with VSWR 1.25.
7. The tube shall be oscillating during this test.
8. With test cavity as per Drawing 177-JAN, the temperature coefficient of the cavity is -11 kHz/°C. The temperature coefficient of the tube shall be considered to be the difference between the characteristic of the cavity alone and the tube with the cavity. Test one tube every 2 weeks when in continuous production.
9. Total current to the interaction gap electrodes shall be adjusted to 20 mA by appropriate adjustment of Ec1 at the beginning of life test. Ec1 shall be changed during life test to maintain total current to the interaction gap electrodes to 20 mA. The maximum bias of 18 volts on grid No. 1 shall be observed.
10. Between g3 and repeller, 2,000 Vdc shall be applied under an atmospheric pressure of 3.2 inches of mercury (corresponding to an altitude of 50,000 feet). No voltage breakdown shall occur.
11. With test cavity as per Drawing 176-JAN. Loop and plug adjusted for maximum power output into a fixed 50-ohm load with VSWR 1.25.
12. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy -
Air Force - 11, 80
DSA - ES

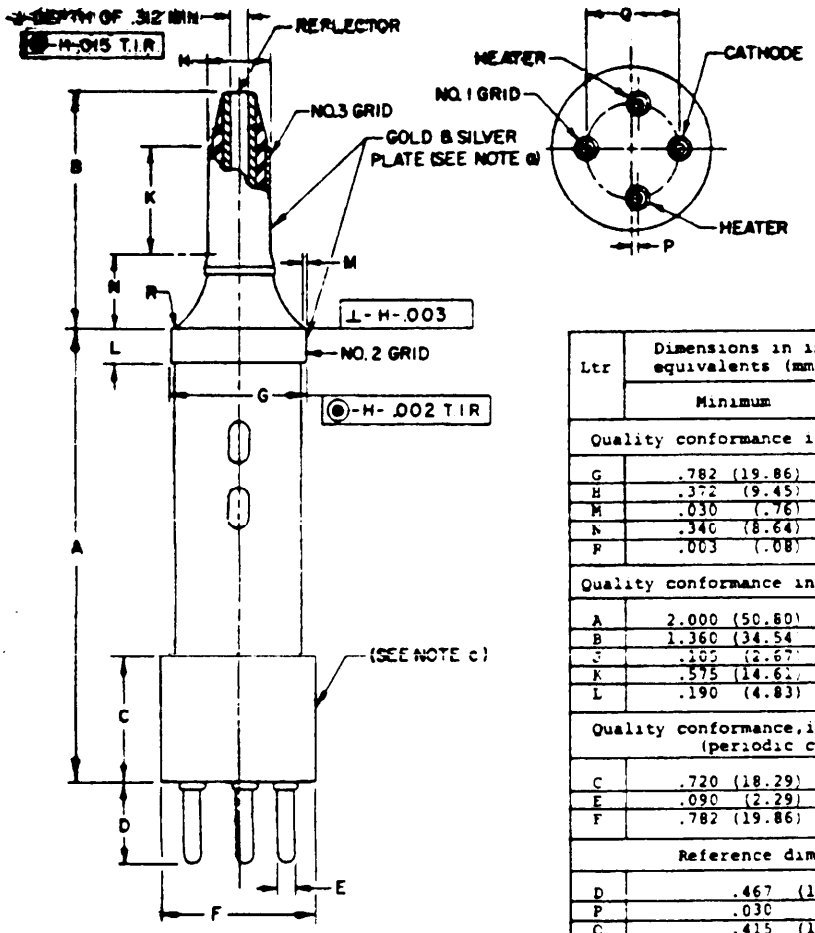
User activities:
Army - WC
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Preparing activity: Navy - EC

Agent: DSA - ES

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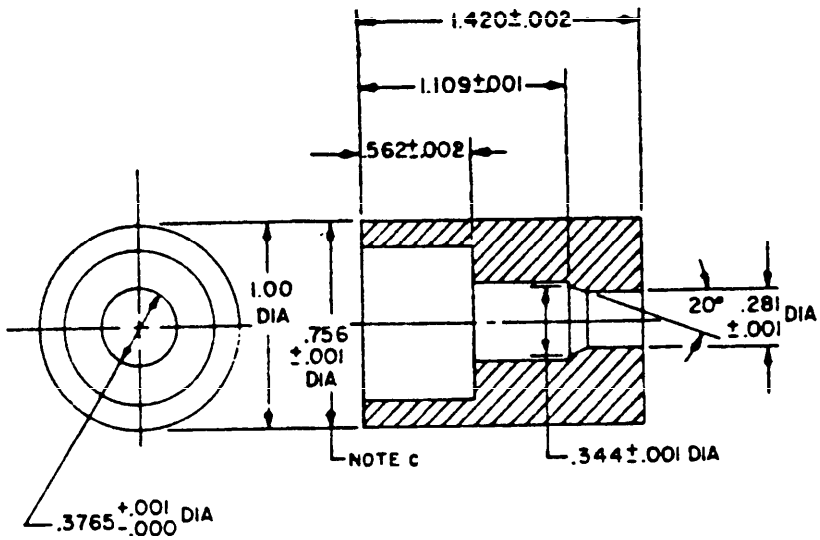


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		NOTES
	Minimum	Maximum	
Quality conformance inspection, part 1			12
G	.782 (19.86)	.786 (19.96)	
H	.372 (9.45)	.376 (9.55)	
M	.030 (.76)		
N	.340 (8.64)	.407 (10.34)	
R	.003 (.08)	.008 (.20)	
Quality conformance inspection, part 2			
A	2.000 (50.80)	2.220 (56.39)	
B	1.360 (34.54	1.420 (36.07)	
[illegible]	.105 (2.67)	.113 (2.87)	
K	.575 (14.61)	.675 (17.15)	
L	.190 (4.83)	.230 (5.84)	
Quality conformance, inspection, part 3 (periodic check)			
C	.720 (18.29)	.780 (19.81)	
E	.090 (2.29)	.097 (2.46)	
F	.782 (19.86)	.844 (21.44)	
Reference dimensions			
D	.467 (11.86)		
P	.030 (.76)		
Q	.415 (10.54)		

NOTES:
a. Plating of grid No. 2 and grid No. 3 shall be a minimum of .00005 inch gold silver. The total plate shall be .0003 inch minimum.
b. Thermal expansion coefficient of grid No. 2 and grid No. 3 shall be such that undue strain does not develop in the glass to metal seal.
c. Low-loss phenolic base.

FIGURE 1. Outline drawing of electron tube type 5721.

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INCHES	MM
.001	.03
.002	.05
.281	7.14
.344	8.74
.3765	9.56
.562	14.27
.756	19.20
1.000	25.40
1.109	28.17
1.420	36.07

NOTES:
a. The internal diameters shall be concentric within .0005(.01 mm) TIR.
b. Tube shall enter far enough to seat on the leading edge of the g2 ring.
c. To largest I.D.

FIGURE 2. Reflector insulator gage.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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